

```
postgres=# SELECT e.ename
FROM employee e
JOIN department d ON e.dept_id = d.dept_id
WHERE d.department_name = 'IT'
ORDER BY e.ename ASC;
         ename
```

```
-----
Deepak Rajbhandari
Medha Joshi
Pooja Khanal
Sunita Chettri
Tika Gurung
(5 rows)
```

```
postgres=# SELECT e.ename,
        d.department_name,
        e.salary
FROM employee e
JOIN department d ON e.dept_id = d.dept_id
WHERE e.salary = (
        SELECT MAX(e2.salary)
        FROM employee e2
        WHERE e2.dept_id = e.dept_id
);
```

```
         ename      | department_name | salary
-----+-----+-----
Sanjay Puri      | Sociology       | 50000.00
Arjun Sharma     | Humanities      | 42000.00
Surya Adhikari   | Computer-Science | 52000.00
Medha Joshi      | IT              | 47000.00
(4 rows)
```

```
postgres=# SELECT d.department_name, COUNT(e.emp_id) AS employee_count
FROM employee e
JOIN department d ON e.dept_id = d.dept_id
WHERE e.salary > 5000
GROUP BY d.department_name;
 department_name | employee_count
```

```
-----+-----
Sociology      |              3
Humanities     |              4
IT             |              5
Computer-Science |              3
(4 rows)
```

```
postgres=# SELECT d.department_name, AVG(e.salary) AS average_salary
FROM employee e
JOIN department d ON e.dept_id = d.dept_id
GROUP BY d.department_name;
 department_name | average_salary
```

```
-----+-----
Humanities      | 33500.000000000000
Sociology       | 45666.666666666667
Computer-Science | 41333.333333333333
IT              | 38200.000000000000
(4 rows)
```

```

postgres=# SELECT e.ename,
                d.department_name
FROM employee e
LEFT JOIN department d ON e.dept_id = d.dept_id;

```

ename	department_name
Ashok Gautam	Computer-Science
Lhakpa Sherpa	Humanities
Sanjay Puri	Sociology
Arjun Sharma	Humanities
Rina Thapa	Computer-Science
Manoj Koirala	Sociology
Surya Adhikari	Computer-Science
Kiran Singh	Humanities
Bikash Rai	Sociology
Nirajan Chhetri	Humanities
Tika Gurung	IT
Deepak Rajbhandari	IT
Sunita Chettri	IT
Medha Joshi	IT
Pooja Khanal	IT

(15 rows)

```

postgres=# SELECT DISTINCT salary
FROM employee
ORDER BY salary DESC
OFFSET 1 LIMIT 1;

```

salary
50000.00

(1 row)

```

postgres=# SELECT j.job_id, j.job_title, COUNT(e.emp_id) AS total_employees
FROM employee e
JOIN job j ON e.job_id = j.job_id
GROUP BY j.job_id, j.job_title
HAVING COUNT(e.emp_id) > 3;

```

job_id	job_title	total_employees
4	IT Manager	5
1	Professor	4

(2 rows)

```

postgres=# SELECT ename, emp_id, job_id, dept_id
FROM employee
WHERE manager_id IS NULL;

```

ename	emp_id	job_id	dept_id
Lhakpa Sherpa	202	1	1
Ashok Gautam	201	2	2
Sanjay Puri	203	3	3
Medha Joshi	213	4	11

(4 rows)

```

postgres=# SELECT e.ename, e.salary, j.job_title
FROM employee e
JOIN job j ON e.job_id = j.job_id
ORDER BY e.salary DESC
LIMIT 3;

```

ename	salary	job_title
Surya Adhikari	52000.00	Lecturer
Sanjay Puri	50000.00	Researcher
Medha Joshi	47000.00	IT Manager

(3 rows)

```

postgres=# SELECT d.department_name, AVG(e.salary) AS avg_salary
FROM employee e
JOIN department d ON e.dept_id = d.dept_id
GROUP BY d.department_name
ORDER BY avg_salary DESC
LIMIT 1;

```

department_name	avg_salary
Sociology	45666.666666666667

(1 row)

```

postgres=# SELECT ename, hire_date
FROM employee
WHERE hire_date > '2005-12-31'
ORDER BY hire_date ASC;

```

ename	hire_date
Ashok Gautam	2006-05-12
Lhakpa Sherpa	2006-05-12
Sanjay Puri	2006-05-12
Manoj Koirala	2008-11-10
Tika Gurung	2008-11-10
Deepak Rajbhandari	2008-11-10
Kiran Singh	2009-07-05
Surya Adhikari	2009-07-05
Bikash Rai	2009-07-05
Sunita Chettri	2009-07-05
Medha Joshi	2010-01-20
Nirajan Chhetri	2010-01-20
Pooja Khanal	2010-01-20

(13 rows)

```

postgres=# SELECT j.job_title, COUNT(e.emp_id) AS total_employees
FROM employee e
JOIN job j ON e.job_id = j.job_id
GROUP BY j.job_title
ORDER BY total_employees DESC;

```

job_title	total_employees
IT Manager	5
Professor	4
Researcher	3
Lecturer	3

(4 rows)

```

postgres=# SELECT e.ename, d.department_name, d.location
FROM employee e
JOIN department d ON e.dept_id = d.dept_id
WHERE d.location ILIKE '%London%';

```

ename	department_name	location
-----+-----+-----		
Steven King	Consulting	London
Sanjay Puri	Sociology	London
Manoj Koirala	Sociology	London
Bikash Rai	Sociology	London
Ashok Gautam	Computer-Science	London
Rina Thapa	Computer-Science	London
Surya Adhikari	Computer-Science	London
(7 rows)		

```

postgres=# SELECT e.ename, j.job_title, d.department_name
FROM employee e
JOIN job j ON e.job_id = j.job_id
JOIN department d ON e.dept_id = d.dept_id
WHERE e.ename = 'Steven King';

```

ename	job_title	department_name
-----+-----+-----		
Steven King	Consultant	Consulting
(1 row)		

```

postgres=# SELECT j.job_title,
                AVG(e.salary) AS avg_salary,
                MIN(e.salary) AS min_salary,
                MAX(e.salary) AS max_salary
FROM employee e
JOIN job j ON e.job_id = j.job_id
GROUP BY j.job_title
ORDER BY avg_salary DESC;

```

job_title	avg_salary	min_salary	max_salary
-----+-----+-----			
Researcher	45666.666666666667	41000.00	50000.00
Lecturer	41333.333333333333	35000.00	52000.00
IT Manager	38200.000000000000	28000.00	47000.00
Professor	33500.000000000000	25000.00	42000.00
(4 rows)			

```

postgres=# CREATE TABLE Student (
    Student_ID SERIAL PRIMARY KEY,
    Name VARCHAR(50),
    Department VARCHAR(50),
    Age INT
);
CREATE TABLE

```

```

postgres=# UPDATE Student
SET Age = 23
WHERE Student_ID = 2;

DELETE FROM Student
WHERE Student_ID = 3;
UPDATE 1
DELETE 1

```

```

postgres=# INSERT INTO Student (Name, Department, Age) VALUES
('Ram', 'CSIT', 20),
('Sita', 'CSIT', 22),
('Hari', 'Math', 21);
INSERT 0 3

```

```

postgres=# SELECT * FROM Student WHERE Department = 'CSIT';
 student_id | name | department | age
-----+-----+-----+-----
          1 | Ram  | CSIT       | 20
          2 | Sita | CSIT       | 22
(2 rows)

```

```

postgres=# SELECT e.*
FROM employee e
JOIN job j ON e.job_id = j.job_id
WHERE j.job_title = 'SH_CLERK';
 emp_id | first_name | last_name | salary | dept_id |      ename      | job_id | manager_id | hire_date
-----+-----+-----+-----+-----+-----+-----+-----+-----
    202 | Lhakpa    | Sherpa   | 25000.00 |      1 | Lhakpa Sherpa   |     17 |           | 2006-05-12
    210 | Kiran     | Singh    | 31000.00 |      1 | Kiran Singh     |     17 |    202    | 2009-07-05
    214 | Nirajan   | Chhetri  | 36000.00 |      1 | Nirajan Chhetri |     17 |    202    | 2010-01-20
(3 rows)

```

```

postgres=# SELECT e.first_name, e.last_name
FROM employee e
JOIN department d ON e.dept_id = d.dept_id
WHERE d.location = (
    SELECT location
    FROM department
    WHERE dept_id = 90
);
 first_name | last_name
-----+-----
    Lhakpa   | Sherpa
    Kiran    | Singh
   Nirajan   | Chhetri
    Arjun    | Sharma
    Ashok    | Gautam
    Rina     | Thapa
    Surya    | Adhikari
    Tika     | Gurung
    Bikash   | Rai
    Sunita   | Chettri
(10 rows)

```

```

postgres=# SELECT d.department_name
FROM employee e
JOIN department d ON e.dept_id = d.dept_id
GROUP BY d.department_name
HAVING COUNT(e.emp_id) = 2;
 department_name
-----
    Sociology
(1 row)

```

```

postgres=# SELECT *
FROM employee
WHERE hire_date > (
    SELECT hire_date
    FROM employee
    WHERE first_name = 'Neena' AND last_name = 'Kochhar'
);
 emp_id | first_name | last_name | salary | dept_id |      ename      | job_id | manager_id | hire_date
-----+-----+-----+-----+-----+-----+-----+-----+-----
    301 | Steven    | King      | 60000.00 |    33 | Steven King      |      6 |           | 2012-03-15
    210 | Kiran     | Singh     | 31000.00 |     1 | Kiran Singh      |     17 |          202 | 2009-07-05
    214 | Nirajan   | Chhetri    | 36000.00 |     1 | Nirajan Chhetri   |     17 |           | 2010-01-20
    207 | Tika      | Gurung     | 28000.00 |    90 | Tika Gurung       |      4 |           | 2008-11-10
    212 | Bikash    | Rai        | 41000.00 |    90 | Bikash Rai        |      3 |           | 2009-07-05
    211 | Sunita    | Chettri    | 33000.00 |    90 | Sunita Chettri    |      4 |           | 2009-07-05
    218 | Michael   | Hartstein  | 48000.00 |     2 | Michael Hartstein |      2 |           | 2008-01-20
    219 | Alice     | Sharma     | 42000.00 |     2 | Alice Sharma      |      2 |           | 2009-04-10
    220 | Bob       | Thapa      | 40000.00 |     2 | Bob Thapa         |      2 |           | 2010-07-05
    206 | Manoj     | Koirala    | 46000.00 |     3 | Manoj Koirala     |      3 |           | 2008-11-10
    208 | Deepak    | Rajbhandari | 39000.00 |    11 | Deepak Rajbhandari |      4 |           | 2008-11-10
    209 | Surya     | Adhikari   | 52000.00 |     2 | Surya Adhikari    |      2 |           | 2009-07-05
    213 | Medha     | Joshi      | 47000.00 |    11 | Medha Joshi       |      4 |           | 2010-01-20
    215 | Pooja     | Khanal     | 44000.00 |    11 | Pooja Khanal      |      4 |          213 | 2010-01-20
(14 rows)

```



```

postgres=# SELECT *
FROM employee
WHERE manager_id = (
    SELECT manager_id
    FROM employee
    WHERE first_name = 'Michael' AND last_name = 'Hartstein'
);

```

emp_id	first_name	last_name	salary	dept_id	ename	job_id	manager_id	hire_date
218	Michael	Hartstein	48000.00	2	Michael Hartstein	2	217	2008-01-20
219	Alice	Sharma	42000.00	2	Alice Sharma	2	217	2009-04-10
220	Bob	Thapa	40000.00	2	Bob Thapa	2	217	2010-07-05

(3 rows)

```

postgres=# -- Create view
CREATE VIEW emp_salaries AS
SELECT emp_id, first_name, last_name, salary
FROM employee;

```

```

-- Display results from the view

```

```

SELECT * FROM emp_salaries;

```

```

CREATE VIEW

```

emp_id	first_name	last_name	salary
301	Steven	King	60000.00
202	Lhakpa	Sherpa	25000.00
210	Kiran	Singh	31000.00
214	Nirajan	Chhetri	36000.00
207	Tika	Gurung	28000.00
212	Bikash	Rai	41000.00
211	Sunita	Chettri	33000.00
216	Neena	Kochhar	55000.00
217	David	Manager	60000.00
218	Michael	Hartstein	48000.00
219	Alice	Sharma	42000.00
220	Bob	Thapa	40000.00
201	Ashok	Gautam	35000.00
203	Sanjay	Puri	50000.00
204	Arjun	Sharma	42000.00
205	Rina	Thapa	37000.00
206	Manoj	Koirala	46000.00
208	Deepak	Rajbhandari	39000.00
209	Surya	Adhikari	52000.00
213	Medha	Joshi	47000.00


```

postgres=# CREATE VIEW high_earners AS
SELECT emp_id, first_name, last_name, salary
FROM employee
WHERE salary > 10000;

```

```

-- Display results
SELECT * FROM high_earners;
CREATE VIEW

```

emp_id	first_name	last_name	salary
301	Steven	King	60000.00
202	Lhakpa	Sherpa	25000.00
210	Kiran	Singh	31000.00
214	Nirajan	Chhetri	36000.00
207	Tika	Gurung	28000.00
212	Bikash	Rai	41000.00
211	Sunita	Chettri	33000.00
216	Neena	Kochhar	55000.00
217	David	Manager	60000.00
218	Michael	Hartstein	48000.00
219	Alice	Sharma	42000.00
220	Bob	Thapa	40000.00
201	Ashok	Gautam	35000.00
203	Sanjay	Puri	50000.00
204	Arjun	Sharma	42000.00
205	Rina	Thapa	37000.00
206	Manoj	Koirala	46000.00
208	Deepak	Rajbhandari	39000.00
209	Surya	Adhikari	52000.00
213	Medha	Joshi	47000.00

```

postgres=# CREATE VIEW emp_dept AS
SELECT e.first_name, e.last_name, d.department_name
FROM employee e
JOIN department d ON e.dept_id = d.dept_id;

```

```

-- Display results
SELECT * FROM emp_dept;
CREATE VIEW

```

first_name	last_name	department_name
Steven	King	Consulting
Lhakpa	Sherpa	Humanities
Kiran	Singh	Humanities
Nirajan	Chhetri	Humanities
Tika	Gurung	Research & Development
Bikash	Rai	Research & Development
Sunita	Chettri	Research & Development
Neena	Kochhar	Humanities
David	Manager	Computer-Science
Michael	Hartstein	Computer-Science

```

Alicpostgres=# SELECT DISTINCT first_name
FROM employee
WHERE dept_id IN (10, 20);

```

```

Sanj first_name
Arju-----
Rina Anita
Mano Binod
Deep Manish
Sury Priya
Medh Raj
      Suman
      (6 rows)

```

```
postgres=# SELECT first_name
FROM employee
WHERE dept_id IN (10, 20);
 first_name
-----
Raj
Suman
Anita
Manish
Priya
Binod
(6 rows)
```

```
postgres=# SELECT first_name
FROM employee
WHERE dept_id = 30
      AND salary > 8000;
 first_name
-----
Ramesh
Sita
(2 rows)
```

```
postgres=# SELECT first_name
FROM employee
WHERE dept_id = 30
      AND salary <= 8000;
 first_name
-----
Hari
(1 row)
```

```
postgres=# SELECT DISTINCT job_id
FROM employee
WHERE dept_id IN (10, 20)
ORDER BY job_id;
 job_id
-----
      2
      3
(2 rows)
```

